

FRM PART I – MOCK I ANSWERS

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Reference Table: Let Z be a standard normal random variable.

z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)
-3	0.0013	-2.50	0.0062	-2.00	0.0228	-1.50	0.0668	-1.00	0.1587	-0.50	0.3085
-2.99	0.0014	-2.49	0.0064	-1.99	0.0233	-1.49	0.0681	-0.99	0.1611	-0.49	0.3121
-2.98	0.0014	-2.48	0.0066	-1.98	0.0239	-1.48	0.0694	-0.98	0.1635	-0.48	0.3156
-2.97	0.0015	-2.47	0.0068	-1.97	0.0244	-1.47	0.0708	-0.97	0.1660	-0.47	0.3192
-2.96	0.0015	-2.46	0.0069	-1.96	0.0250	-1.46	0.0721	-0.96	0.1685	-0.46	0.3228
-2.95	0.0016	-2.45	0.0071	-1.95	0.0256	-1.45	0.0735	-0.95	0.1711	-0.45	0.3264
-2.94	0.0016	-2.44	0.0073	-1.94	0.0262	-1.44	0.0749	-0.94	0.1736	-0.44	0.3300
-2.93	0.0017	-2.43	0.0075	-1.93	0.0268	-1.43	0.0764	-0.93	0.1762	-0.43	0.3336
-2.92	0.0018	-2.42	0.0078	-1.92	0.0274	-1.42	0.0778	-0.92	0.1788	-0.42	0.3372
-2.91	0.0018	-2.41	0.0080	-1.91	0.0281	-1.41	0.0793	-0.91	0.1814	-0.41	0.3409
-2.9	0.0019	-2.40	0.0082	-1.90	0.0287	-1.40	0.0808	-0.90	0.1841	-0.40	0.3446
-2.89	0.0019	-2.39	0.0084	-1.89	0.0294	-1.39	0.0823	-0.89	0.1867	-0.39	0.3483
-2.88	0.0020	-2.38	0.0087	-1.88	0.0301	-1.38	0.0838	-0.88	0.1894	-0.38	0.3520
-2.87	0.0021	-2.37	0.0089	-1.87	0.0307	-1.37	0.0853	-0.87	0.1922	-0.37	0.3557
-2.86	0.0021	-2.36	0.0091	-1.86	0.0314	-1.36	0.0869	-0.86	0.1949	-0.36	0.3594
-2.85	0.0022	-2.35	0.0094	-1.85	0.0322	-1.35	0.0885	-0.85	0.1977	-0.35	0.3632
-2.84	0.0023	-2.34	0.0096	-1.84	0.0329	-1.34	0.0901	-0.84	0.2005	-0.34	0.3669
-2.83	0.0023	-2.33	0.0099	-1.83	0.0336	-1.33	0.0918	-0.83	0.2033	-0.33	0.3707
-2.82	0.0024	-2.32	0.0102	-1.82	0.0344	-1.32	0.0934	-0.82	0.2061	-0.32	0.3745
-2.81	0.0025	-2.31	0.0104	-1.81	0.0351	-1.31	0.0951	-0.81	0.2090	-0.31	0.3783
-2.8	0.0026	-2.30	0.0107	-1.80	0.0359	-1.30	0.0968	-0.80	0.2119	-0.30	0.3821
-2.79	0.0026	-2.29	0.0110	-1.79	0.0367	-1.29	0.0985	-0.79	0.2148	-0.29	0.3859
-2.78	0.0027	-2.28	0.0113	-1.78	0.0375	-1.28	0.1003	-0.78	0.2177	-0.28	0.3897
-2.77	0.0028	-2.27	0.0116	-1.77	0.0384	-1.27	0.1020	-0.77	0.2206	-0.27	0.3936
-2.76	0.0029	-2.26	0.0119	-1.76	0.0392	-1.26	0.1038	-0.76	0.2236	-0.26	0.3974
-2.75	0.0030	-2.25	0.0122	-1.75	0.0401	-1.25	0.1056	-0.75	0.2266	-0.25	0.4013
-2.74	0.0031	-2.24	0.0125	-1.74	0.0409	-1.24	0.1075	-0.74	0.2296	-0.24	0.4052
-2.73	0.0032	-2.23	0.0129	-1.73	0.0418	-1.23	0.1093	-0.73	0.2327	-0.23	0.4090
-2.72	0.0033	-2.22	0.0132	-1.72	0.0427	-1.22	0.1112	-0.72	0.2358	-0.22	0.4129
-2.71	0.0034	-2.21	0.0136	-1.71	0.0436	-1.21	0.1131	-0.71	0.2389	-0.21	0.4168
-2.7	0.0035	-2.20	0.0139	-1.70	0.0446	-1.20	0.1151	-0.70	0.2420	-0.20	0.4207
-2.69	0.0036	-2.19	0.0143	-1.69	0.0455	-1.19	0.1170	-0.69	0.2451	-0.19	0.4247
-2.68	0.0037	-2.18	0.0146	-1.68	0.0465	-1.18	0.1190	-0.68	0.2483	-0.18	0.4286
-2.67	0.0038	-2.17	0.0150	-1.67	0.0475	-1.17	0.1210	-0.67	0.2514	-0.17	0.4325
-2.66	0.0039	-2.16	0.0154	-1.66	0.0485	-1.16	0.1230	-0.66	0.2546	-0.16	0.4364
-2.65	0.0040	-2.15	0.0158	-1.65	0.0495	-1.15	0.1251	-0.65	0.2578	-0.15	0.4404
-2.64	0.0041	-2.14	0.0162	-1.64	0.0505	-1.14	0.1271	-0.64	0.2611	-0.14	0.4443
-2.63	0.0043	-2.13	0.0166	-1.63	0.0516	-1.13	0.1292	-0.63	0.2643	-0.13	0.4483
-2.62	0.0044	-2.12	0.0170	-1.62	0.0526	-1.12	0.1314	-0.62	0.2676	-0.12	0.4522
-2.61	0.0045	-2.11	0.0174	-1.61	0.0537	-1.11	0.1335	-0.61	0.2709	-0.11	0.4562
-2.6	0.0047	-2.10	0.0179	-1.60	0.0548	-1.10	0.1357	-0.60	0.2743	-0.10	0.4602
-2.59	0.0048	-2.09	0.0183	-1.59	0.0559	-1.09	0.1379	-0.59	0.2776	-0.09	0.4641
-2.58	0.0049	-2.08	0.0188	-1.58	0.0571	-1.08	0.1401	-0.58	0.2810	-0.08	0.4681
-2.57	0.0051	-2.07	0.0192	-1.57	0.0582	-1.07	0.1423	-0.57	0.2843	-0.07	0.4721
-2.56	0.0052	-2.06	0.0197	-1.56	0.0594	-1.06	0.1446	-0.56	0.2877	-0.06	0.4761
-2.55	0.0054	-2.05	0.0202	-1.55	0.0606	-1.05	0.1469	-0.55	0.2912	-0.05	0.4801
-2.54	0.0055	-2.04	0.0207	-1.54	0.0618	-1.04	0.1492	-0.54	0.2946	-0.04	0.4840
-2.53	0.0057	-2.03	0.0212	-1.53	0.0630	-1.03	0.1515	-0.53	0.2981	-0.03	0.4880
-2.52	0.0059	-2.02	0.0217	-1.52	0.0643	-1.02	0.1539	-0.52	0.3015	-0.02	0.4920
-2.51	0.0060	-2.01	0.0222	-1.51	0.0655	-1.01	0.1562	-0.51	0.3050	-0.01	0.4960

Reference Table: t-table.

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

FRM Part I – Mock Exam II –Candidate Answer Sheet

1.		26.		51.		76.	
2.		27.		52.		77.	
3.		28.		53.		78.	
4.		29.		54.		79.	
5.		30.		55.		80.	
6.		31.		56.		81.	
7.		32.		57.		82.	
8.		33.		58.		83.	
9.		34.		59.		84.	
10.		35.		60.		85.	
11.		36.		61.		86.	
12.		37.		62.		87.	
13.		38.		63.		88.	
14.		39.		64.		89.	
15.		40.		65.		90.	
16.		41.		66.		91.	
17.		42.		67.		92.	
18.		43.		68.		93.	
19.		44.		69.		94.	
20.		45.		70.		95.	
21.		46.		71.		96.	
22.		47.		72.		97.	
23.		48.		73.		98.	
24.		49.		74.		99.	
25.		50.		75.		100.	

Foundations of Risk Management

1. Hannah Zane is a fixed-income portfolio manager at Smart Capital. She comes across a research report issued by a major brokerage firm on Fortunes Corporation in which the research analyst argues that within one year, the Fortune's credit fundamentals will strengthen. The report goes further to assert following the narrowing spread, the market will respond by demanding a lower credit spread. After reading the report, Ms. Zane decides that she wants to take a credit view on Fortunes Corporation.

Next week, Fortunes Corporation has declared its intention to bring to the market a 10-year senior bond issue at par with a coupon rate of 12%, offering a spread of 800 basis points over the corresponding 10-year Treasury issue. Ms. Zane is not keen to purchase the bond outright because she does not want to bear the out-of-pocket costs, the inconvenience of arranging the financing, actually going long the bond and taking delivery. Instead, she would like to express her view on Fortune Corporation's credit risk by entering into a total return swap that matures in one year with the senior bonds that are about to be issued as the reference obligation. Under the terms of the contract, payments will be exchanged semiannually, where the total return receiver will pay the six-month Treasury rate plus 350 basis points.

Which of the following is most likely correct?

- A. Ms. Zane's counterparty will enter the total return swap as the total return receiver.
- B. Ms. Zane will enter the total return swap as the total return receiver.
- C. The total return receiver will be exposed to credit risk but not interest rate risk.
- D. The party that agrees to receive the floating payments and pay the total return is the total return receiver.

2. The macroeconomic research department of a large investment bank uses the arbitrage pricing theory to set up its expectations on equity market performance in the next quarter. It uses the two-factor model based on the short-term interest rate level and the GDP growth rate. The risk-free rate is currently 1%.

	Factor Beta	Factor Risk Premium
Short-term interest rate	-0.9	3%
GDP	1.3	7%

Although market participants do not expect any changes in interest rate, analysts at the bank are strongly confident that there will be an increase of 100 basis points. They also believe the GDP growth rate will beat market expectations by 50 basis points.

Taking into account the expectations of the macroeconomic research department, what is the expected return on the equity market?

- A. 7.15%
- B. 7.25%
- C. 7.40%
- D. 8.30%

3. Which of the following cases include a violation of GARP's rules of conduct?

- A. Frank Benton - a risk manager and participant in the FRM program - oversees all investments in the healthcare startups for an investment bank. After thorough analysis and evaluating all necessary disclosures, Benton approves a significant loan facility to LLL Medicine (a young startup initiated by Frank's brother).
- B. Andrea Jenkins, FRM, is a rising star at Trust Hedge Fund. Behind Jenkins' success is her ability to fully concentrate on investment analysis since all other concerns – like the evaluation of an investment's suitability based on funds' internal rules, verification of compliance with laws, etc. – are handled by a third-party accounting firm.
- C. Elisabeth Stone, FRM, is responsible for credit risk analysis of corporate borrowers in one of Mississauga's local banks. On a quarterly basis, she presents to the board of directors a summary of the key ratios used for identifying the credit quality of existing borrowers with large exposures. The analysis is based on quarterly reports published by the clients. In addition, she always includes one slide with her own opinion on changes in ratios presented in case of adverse market movements. The slide has a footnote clarifying that the presented information is based on Stone's estimations.
- D. Jack Merton, who recently passed part one of the FRM exam, was reviewing his analysis on interest rate risk presented to the ALCO committee of Warsaw Bank last week. He noticed a major mistake and immediately informed all parties involved in this mistake.

4. Which of the following is (are) not (an) assumption(s) of the capital pricing model?

- I. Investors require higher returns as a pre-condition for higher risks
 - II. All investors have the same one-period time horizon
 - III. CAPM considers taxes but does not consider transaction costs
 - IV. Investors' only concern should be a systematic risk
- A. II only
 - B. III only
 - C. I and IV only
 - D. III and IV only

5. Which of the following are the roles of the audit committee?

- I. Members must have the knowledge of basic accounting principles under US GAAP and IFRS
- II. Ensuring that the remuneration of key management aligns with the goals of other stakeholders
- III. Ensuring reasonable accuracy of the organization's financial statements
- IV. Ensuring that management strategies are focused on economic profits
- V. Ensuring that the organization has taken measures to avoid misstated errors and fraud

- A. I, II and IV
- B. III, IV, and V
- C. I, III and V
- D. II, III and IV

6. Which of the following statements inappropriately defines the elements of Enterprise Risk Management (ERM)?

- I. Stakeholders management refers to sharing or communicating the information regarding a firm's internal risk management processes to external stakeholders.
- II. Data technology and resources refer to the management of activities that are directly related to the production of products and services of the firm.

- A. I only
- B. II only
- C. Both I & II
- D. None of the above

7. Enterprise-wide VaR is not likely to account for all types of risk. In particular, enterprise-wide VaR may not factor in:

- A. Credit risk
- B. Market risk
- C. Funding liquidity risk
- D. None of the above. Enterprise-wide risk is likely to account for all types of risk.

8. There are important similarities between the 2007-2009 crisis and the “The Big Five Crisis” of Spain, Norway, Finland, Sweden, and Japan. In all those crises, there were:

I. Notable increases in public debt

II. Positive current account balances

III. Excessive borrowing from foreign institutions and governments

IV. House price run-ups

- A. I and IV
- B. I, III and IV
- C. II, III and IV
- D. All of the above

9. Malala Pham is an equity analyst. She has been asked to derive the beta of a stock from the CAPM. What is the value of the beta if the risk-free rate is 3%, the expected return of the market is 14%, and the return on the stock is 11.8%?

- A. 0.63
- B. 1.5
- C. 0.8
- D. 1.07

10. A stock is trading at USD 35.12, and analysts expect its price in exactly 12 months to be USD 38.77. If the market risk is 8%, the risk-free rate is 5%, and the Beta of the stock is 1.8, the stock is most likely:

- A. Undervalued by USD 3.16
- B. Undervalued by USD 6.81
- C. Undervalued by USD 3.16
- D. Fairly valued

11. The exhibit below summarizes risk and return for three market-neutral hedge funds in Canada:

Exhibit
Risk, Return and Fee Data for Three Market-Neutral Hedge Funds

	Fund A	Fund B	Fund C
Annualized return	15%	22%	9%
Annualized standard deviation	20%	36%	15%

If the risk-free rate in Canada is 2%, which of these funds is most appropriate from an investment perspective, according to the Sharpe ratio?

- A. Fund A
- B. Fund B
- C. Fund C
- D. Insufficient information

12. Which of the following risk metrics measures the risk as downside deviation instead of standard deviation?

- A. Treynor ratio
- B. Sharp ratio
- C. Sortino ratio
- D. Jensen measure

13. The common stock of General Electric is examined with a single factor model using unexpected percentage changes in GDP as the single factor. You have been provided with the following data:

- Expected return for General Electric = 7%
- GDP factor-beta = 0.8
- Expected GDP growth = 4%

Revised macroeconomic information strongly suggests that the GDP will grow by 2% as opposed to the original prediction of 4%. Assuming there's no new information regarding firm-specific events, the revised expected return using a single factor model is closest to:

- A. 5.9%
- B. 6.2%
- C. 8.6%
- D. 5.4%

14. Consider a single factor APT in a country where the risk-free rate is 8%. The stocks of XENA have a beta of 1.7 and an expected return of 19%. The stocks of YNN have a beta of 2.2 and an expected return of 15%. Assuming you wanted to exploit an arbitrage opportunity, you would take a short position in:

- A. YNN and use the proceeds to take a long position in XENA
- B. XENA and use the proceeds to take a long position in YNN
- C. YNN and use the proceeds to take a long position in the riskless asset
- D. XENA and use the proceeds to take a long position in the riskless asset

15. Jared Bosh, FRM, serves as a portfolio manager at Cali Bank. Among his clients is Gustavo Hernandez, a talented chef and restaurant owner, for whom Bosh manages a personal portfolio of stocks and government bonds.

Hernandez recently disclosed to Bosh that he is under investigation for being an illegal immigrant and might get deported in the coming months. Later that day, Bosh shares that information with his colleague at Cali Bank responsible for handling the business loans of Hernandez.

Jared Bosh has most likely:

- A. Not violated the GARP Code of Conduct because she revealed illegal activities on the part of her client
- B. Not violated the GARP Code of Conduct with respect to confidentiality because it is central to the bank's activities to know that a customer with outstanding loans might get deported
- C. Violated the GARP Code of Conduct by failing to detect the immigration status of his client
- D. Violated the GARP Code of Conduct with respect to confidentiality

16. Which of the following is not included in the GARP Rules of Conduct?

GARP members shall:

- A. Act professionally, ethically and with integrity in all dealings with employers, existing or potential clients, the public, and other practitioners in the financial services industry.
- B. Not knowingly misrepresent details relating to analysis, recommendations, actions, or other professional activities.
- C. Not make use of confidential information for inappropriate purposes and unless having received prior consent shall maintain the confidentiality of their work, their employer or client.
- D. Act with honesty, integrity, and competence to fulfill the risk professional's responsibilities and to uphold the reputation of the risk management profession.

17. Xiao Bing, head of compliance at OceanAsian Bank, quoted a key governance principle related to risk data aggregation provided by the Basel Committee, which states that “a bank should be able to generate aggregate risk data to meet a broad range of on-demand, ad hoc risk management reporting requests, including requests during stress/crisis situations, requests due to changing internal needs and requests to meet supervisory queries”.

Which of the following principles is Bing referring to?

- A. Accuracy and Integrity
- B. Completeness
- C. Adaptability
- D. Timeliness

18. A 5-year research has been conducted on 25 different hedge funds in the United States. The researchers have categorized the funds into 4 distinct hedge fund strategies:

Fund strategy	Average annual return	Standard deviation	Beta
Global Macro	13.5%	18%	1.15
Long-Short Equity	21.3%	31%	3.1
Equity arbitrage	22.6%	35%	3.4
Fixed income arbitrage	8%	2%	0.6

If the risk-free rate is 4% and the S&P 500 has returned an average of 10.1% over the 5 year period, then which fund strategy has performed the best according to the Treynor measure?

- A. Global Macro
- B. Long-Short Equity
- C. Equity arbitrage
- D. Fixed income arbitrage

19. Which of the following best defines a nonrecourse loan?

- A. A type of loan secured by collateral in which the issuer can seize the collateral but cannot seek from the borrower for any further compensation if the borrower defaults
- B. A type of unsecured loan in which the issuer cannot seek out the borrower for any compensation if the borrower defaults
- C. A type of loan secured by collateral, usually a property, in which the issuer can only seek further compensation in the event the collateral does not cover the full value of the defaulted amount
- D. A type of unsecured loan in which the borrower pays a very low initial interest **rate** that increases after a few years

20. According to the Capital Asset Pricing Model (CAPM), which of the following risks should an investor be compensated for?

I. Unsystematic risk

II. Asset-specific risk

III. Systematic risk

- A. I only
- B. III only
- C. I and II only
- D. I and III only

Quantitative Analysis

21. An analyst is performing a regression analysis on the daily prices of Google (GOOG) compared to the prices of the S&P 500 index. Below is the data that the analyst managed to collect for the last 5 trading days:

Closing Price	GOOG	S&P
Dec 05, 2017	1,005	2,630
Dec 04, 2017	999	2,639
Dec 01, 2017	1,010	2,642
Nov 30, 2017	1,021	2,648
Nov 29, 2017	1,022	2,626

What is the equation for the regression line?

- A. $y = 995 + 0.00625x$
 - B. $y = 574 + 0.859x$
 - C. $y = 0.859 + 574x$
 - D. $y = 989 + 0.859x$
22. Luke Friday, FRM, runs a consultancy firm that offers investment advice to clients around Los Angeles. The number of clients the firm receives in a month is distributed as a Poisson variable with a mean of 2. What is the probability that the firm receives exactly 30 new clients in a year, assuming every client is independent?
- A. 0.025
 - B. 0.0363
 - C. 0.24
 - D. 0.00363
23. A portfolio manager has recently invested in two stocks – HKJ and YYG. The probability that HKJ posts a return greater than 10% this year is 0.35, and the probability that YYG posts a return greater than 10% is 0.45.
- Given that the probability that both stocks post returns greater than 10% this year is 0.15, what is the probability that either HKJ or the YYG or both stocks post returns greater than 10% this year?
- A. 0.75
 - B. 0.55
 - C. 0.95
 - D. 0.65

24. A retail investor has recently bought two stocks – UUK and LLY – which have a correlation of 0.72. The covariance between UUK and LLY is 0.032.

Given that LLY's return has a standard deviation of 0.17, the variance of returns for UUK is closest to:

- A. 0.13
- B. 0.06835
- C. 0.0385
- D. 0.0148

25. The FRM part 1 exam attracts candidates from all over the world. A quantitative analyst compiled the following statistics:

Country of residency	Probability of passing the FRM Part 1 exam	A portion of total candidates
United States	0.40	0.22
India	0.42	0.21
Canada	0.39	0.14
Other	0.31	0.44

If a randomly selected candidate has just passed the FRM part 1 exam, the probability that the candidate comes from India is closest to:

- A. 22%
- B. 24%
- C. 25%
- D. 9%

26. A random sample of 120 portfolio managers in the United States was found to have an average salary of USD 72,000. The standard deviation among portfolio managers is known as approximately USD 7,800.

Assuming that salaries follow a normal distribution, what is the 2-sided 95% confidence interval for the mean salary of FRM portfolio managers in the United States?

- A. (USD 70,162.94, USD 73,837.06)
- B. (USD 56,712, USD 87,288)
- C. (USD 71,872.60, USD 72,127.40)
- D. (USD 70,604.40, USD 73,395.60)

27. A podiatrist has the following data regarding the height of pre-teenagers:

Age in years	11	12	13
Height in centimeters	144	150	157

Further information: $S_{XX} = 66$, $S_{YY} = 3.11$, $S_{XY} = 11.3$

The Y-intercept is closest to:

- A. 148.27
- B. 106.73
- C. -104.33
- D. 80.24

28. The estimated slope coefficient (β_1) for a certain stock is 0.6566, with a standard error equal to 0.0799. The sample had 10 observations, and a researcher wants to know if the slope coefficient is statistically different than zero. Using a 5% level of significance, what are the test statistic and the decision rule?

- A. 0.1217; do not reject H_0
- B. 8.2177; do not reject H_0
- C. 0.1217; reject H_0
- D. 8.2177; reject H_0

29. Which of the following are assumptions of the multiple linear regression model?

- I. The error term has an expected value equal to zero and is normally distributed
- II. The error for one observation is uncorrelated with that of a different observation
- III. A non-linear relationship exists between the dependent and independent variables
- IV. The independent variables are random, and there is an exact linear relationship between any two or more independent variables

- A. I and II
- B. I and III
- C. II and IV
- D. III and IV

30. An analyst believes that future 15-year real earnings of the S&P 500 are a function of the trailing dividend payout ratio of the stocks in the index (*DB*) and the yield curve slope (*YC*). She collects data and obtains the following multiple regression results:

	Coefficient	Standard error
Intercept	-15.2%	3.589%
DB	0.37	0.099
YC	0.18	0.133

If the number of observations is assumed to be 1003, test the statistical significance of the independent variable *DB* at the 1% level of significance, quoting the value of the test statistic and the conclusion.

(If needed, refer to the t-table at the end of the document.)

- A. Test statistic = 0.2676; The DB regression coefficient is statistically different from zero
- B. Test statistic = 3.7373; The DB regression coefficient is statistically different from zero
- C. Test statistic = 0.2676; The DB regression coefficient is not statistically different from zero
- D. Test statistic = 3.7373; The DB regression coefficient is not statistically different from zero

31. Suppose we have the following random variables:

{12 13 54 56 25}

Determine the skewness and kurtosis of the sample data.

- A. Kurtosis=0.7861, Skewness=0.1835
- B. Kurtosis=0.1861, Skewness=0.1865
- C. Kurtosis=0.7661, Skewness=0.1885
- D. Kurtosis=0.8861, Skewness=0.1855

32. Caterpillar sales were USD 45.5 billion for the year 2017. The following table presents the sales for each of the four quarters of 2017:

Quarter	Q1	Q2	Q3	Q4
Sales (in billion USD)	15.2	17.2	8.3	4.8

Which of the following is the seasonal index for Q2 of 2017?

- A. 1.38
- B. 0.38
- C. 0.51
- D. 1.51

33. The following sample autocorrelation estimates are obtained using 333 data points:

Lag	1	2	3
Coefficient	0.12	-0.17	-0.09

The value of the Ljung Box Q-statistic is closest to:

- A. 17.32
- B. -47.40
- C. 0.05
- D. 17.12

34. The estimated standard error for a Monte Carlo simulation without antithetic variables is 3.65. The antithetic variables are now included so that the correlation between the pairs is -0.65 and the simulation is repeated 100 times. What is the percentage change in standard error?

- A. 45.65%
- B. -46.65%
- C. 40.84%
- D. -40.84%

35. A model is estimated as:

$$Y_i = 3 + 1.5X_{1i} - 2X_{2i} + \epsilon_i$$

Where $\sigma_{X_1}^2 = 25$ and $\sigma_{X_2}^2 = 36$.

What is the value of $\rho_{X_1X_2}$ given that $\widehat{\beta_1} = 1.7$ such that the model is reduced to:

$$Y_i = \alpha + \hat{\beta}X_{1i} + \epsilon_i$$

- A. 0.765
- B. -0.0754
- C. -0.635
- D. -0.0833

36. The returns from a portfolio are as shown in the table below.

Time	Price
0	100
1	98.56
2	98.65
3	97.76
4	96.50
5	100.56

What is the value of continuously compounded return at time 3?

- A. 0.81%
- B. -0.81%
- C. 0.91%
- D. -0.91%

37. A CNBC analyst is estimating the future value of cryptocurrency wallets, where the mean ending value in USD is USD 95,000, the standard deviation is USD 19,300, and the number of simulations is 1000. Which of the following is closest to the 95% confidence interval he would find with this simulation?

- A. (USD 93,802.55, USD 96,197.45)
- B. (91,213.34, USD 98,786.66)
- C. (USD 94,962.13, USD 95,037.87)
- D. (USD 94,727.43, USD 95,272.57)

38. Which of the following statements is/are accurate?

The autoregressive moving average (ARMA) process forms an important part of time series analysis since it:

I. Captures a very robust picture of the variable being estimated thanks to the exclusion of lagged random shocks and lagged observations

II. Combines the lagged unobservable random shock characteristic of the MA process with the observed lagged time series characteristic of the AR process

III. Involves gradually-decaying autocorrelations

- A. I & II only
- B. II & III only
- C. I & III only
- D. All of the above

39. The CEO of AlphaTrynn Hedge Fund estimates that the effect of increasing the number of qualified financial analysts hired by one will improve the fund's annual return by 0.3% with a standard error of 0.12%. Assuming the fund's returns are normally distributed, which of the following best represents a 90% 2-sided confidence interval for the size of the slope coefficient?

- A. (0.12%, 0.45%)
- B. (0.27%, 0.61%)
- C. (0.20%, 0.54%)
- D. (0.10%, 0.50%)

40. The world health organization tracked a group of 50,000 Indians for 20 years. At the commencement of the study, 75% were categorized as omnivores, 15% as vegetarians, and 10% as vegans. Results revealed that vegetarians were twice as likely as vegans to die from colon cancer, but only half as likely as omnivores. During the period, a randomly selected group member passed on from colon cancer.

What is the probability that the individual who died was an omnivore?

- A. 88.23%
- B. 75%
- C. 83.33%
- D. 85.56%

Financial Markets and Products

41. An exotic option where the payoff of the option is not based on the price of the underlying at the end of the contract, but on the arithmetic average of the prices during the life of the option is most likely called a/an:

- A. Gap option
- B. Lookback option
- C. Asian option
- D. Compound option

42. Karsley Bank, located in Arizona, wishes to establish its footing in Delaware by acquiring Quota Bank. Employees and senior management of the latter do not want their business to be enjoined to that of the former in part because they feel their bank's future is bright. In particular, the directors of Quota Bank fear that the new owners may opt to fire them in favor of new management located in Arizona. The directors decide to seek advice from a reputable investment bank on how to fend off the takeover.

Assuming you were one of the advisory panel members, which of the following would likely **not** form part of your advice?

- A. That Quota Bank adds to its charter a provision that if another company acquires one-third of the shares, other shareholders have the right to sell their shares to that company for a large premium over market prices
- B. That Quota Bank should file a lawsuit to dispute the possible takeover
- C. That Quota Bank adds to its charter a provision making it impossible for any new owners to terminate the contracts of existing directors
- D. That Quota Bank grants its employees stock options that can be exercised in the event of a takeover

43. The following data gives the mortality expectancy among women in India in 1954.

Age in years	Probability of death within one year	Life expectancy
30	0.001312	47.52
31	0.001345	46.59
32	0.001383	45.65
33	0.001415	44.73

Which of the following is the probability of a woman who is aged 30 dying between ages 31 and 32?

- A. 0.001312
- B. 0.001345
- C. 0.001343
- D. 0.001348

44. Consider the following statements regarding closed-end mutual funds:

- I. The funds offer investors professional management
- II. Shares at times trade at a discount to the NAV
- III. Shares at times trade at a premium to the NAV
- IV. Funds are redeemed at their NAVs

Which of the statements above is/are incorrect?

- A. IV only
- B. II and III only
- C. II, III and IV
- D. All of the above

45. Blackoil Inc. is an American oil producing company that regularly sells oil futures to reduce the risk of fluctuating oil prices. This activity can be described as:

- A. Speculating
- B. Hedging
- C. Clearing
- D. None of the above

46. Fatima Dow, a derivatives trader at HEC Hedge Fund, entered into a March silver futures contract on the New York Mercantile Exchange (NYMEX) to purchase 5,000 troy ounces of silver at the futures price of USD 16.96 per ounce. According to NYMEX rules, the initial margin of USD 6,400 and a maintenance margin of USD 3,000 is required to enter and retain the futures position.

If the price of silver futures contracts dropped to USD 16.92, USD 16.90, and USD 16.73, at the end of the first, second and third day, respectively, then what is the margin account balance at the end of the second day?

- A. USD 5,750
- B. USD 1,850
- C. USD 6,100
- D. USD 2,700

47. Christina Forst, an investment manager, has constructed a portfolio that somewhat mirrors the Nasdaq composite index. The investment manager intends to hedge the portfolio by taking a short position in Nasdaq futures. The current worth of the portfolio is USD 633 million, and the Nasdaq index futures price is 2,906, with each contract on 250 times the index. If the beta of the portfolio is 0.98, then the number of contracts Lange should short to hedge her portfolio is closest to:

- A. 871 futures contracts
- B. 854 futures contracts
- C. 217,825 futures contracts
- D. 213,469 futures contracts

48. You have been provided the following table of zero spot rates:

Year	Zero rates (p.a.)
1	5.6%
2	6.0%
3	8.4%
4	9.8%

Using the information provided in the table, which of the following is the accurate 1-year forward rate 2 years from now?

- A. 12.6%
- B. 15.6%
- C. 10.8%
- D. 13.2%

49. Blake Young is a derivative investment manager at Yankee Bank based in New York City. He manages a portfolio of fixed income assets and interest rate futures. He currently has a short position in a futures contract on GILTS, the U.K. equivalent to U.S. Treasury securities. As the delivery month is approaching, Young's manager has to choose the cheapest-to-deliver bond from the four available bonds.

If the last settlement price is 93-08, which of the following bond is the cheapest-to-deliver?

Bond	Quoted Bond Price	Conversion Factor
A	USD 102	1.018
B	USD 103	1.071
C	USD 103	1.089
D	USD 104.50	1.062

- A. Bond A
- B. Bond B
- C. Bond C
- D. Bond D

50. A retail trader is considering borrowing 1000 Euros and investing the proceeds in 1.50-year currency futures contracts on the U.S. dollar. The 1.5-year risk-free interest rates in the Eurozone and the U.S. are 1.36% and 1.55%, respectively, and the spot exchange rate is 1.071 USD per Euro. According to the given data, the futures exchange rate should be 1.074 USD per Euro. However, the 1.5-year futures exchange rate is quoted at 1.100 USD per Euro.

Using this information outlined above, which of the following best represents the arbitrage gain or loss from the strategy?

- A. Arbitrage gain of USD 26.48
- B. Arbitrage loss of USD 26.48
- C. Arbitrage gain of USD 2.06
- D. Arbitrage loss of USD 2.06

51. Tiara Enterprises (TIEN) has just announced its plans to establish a facility in New York, USA, to meet the increased demand for its products. TIEN plans to fund the expansion with debt, and in order to hedge the risk of borrowing, TIEN has entered into a plain vanilla interest rate swap with a notional principal of USD 50 million. TIEN would make semiannual payments at the rate of 12% with the counterparty making floating rate payments at the Euribor rate.

Assuming a 360-day year, if the Euribor was 13.5% on the last settlement date and is 11.0% on the current settlement date, the amount that TIEN would receive at the current settlement date is *closest to*:

- A. USD 250,000.
- B. USD 625,000.
- C. USD 375,000.
- D. USD 312,500

52. James Porter is analyzing in the following option contracts on a security with a current market price of USD 20:

Exhibit: Option Contracts

Call Options	Exercise Price	Time to Expiration
A	USD 25	36 months
B	USD 29	8 months
C	USD 30	22 months

Which of the above option contracts is *more likely* to have the lowest premium?

- A. Call A
- B. Call B
- C. Call C
- D. Either Call A or Call B

53. A fund manager has realized that there is a great potential for profits in the options market without tying up much capital. To test the potential of options trading, he implemented a spread strategy by purchasing a 6-month European call with the strike price of USD 92 and a 6-month European put option with a strike price of USD 84 on the stock of XTR.

Which of the following strategies is the fund manager most likely using?

- A. Straddle strategy
- B. Strip strategy
- C. Strap strategy
- D. Strangle strategy

54. A trader is interested in a strategy in which she purchases an option with the strike price of USD 92 that expires in April. She then invests in another option that starts in April and expires in July. This option will have a strike price equal to the price of the underlying in April. The series of such options is called a:

- A. Cliquet option
- B. Gap option
- C. Forward start option
- D. Spiral futures option

55. Jon Pham, a commodities trader, has analyzed the forward prices of crude oil contracts and realized that there might be an arbitrage profit present in oil futures contracts. The spot price for one barrel of crude oil is USD 62.57, and the 6-month futures contract is quoted as USD 65.35 per barrel. If the risk-free rate is 5%, then the arbitrage profit for trading one crude oil futures contract is closest to:

- A. USD 0
- B. USD 1.20
- C. USD 0.35
- D. USD 0.42

56. Barbara Stapleton has recently joined Ace Investments as an investment manager. Her new boss told her that she would only be trading derivatives on exchanges, and the firm does not approve the use of over-the-counter derivatives. Which of the following derivative instruments is she allowed to trade?

I. Forwards

II. Options

III. Swaps

IV. Futures

- A. I and II only
- B. II and IV only
- C. I, II and IV only
- D. All of the above

57. Which of the following statements is (are) incorrect regarding central counterparties (CCP)?

I. The margining activity of the central counterparty decreases risk, but in some cases, it can also increase risk

II. The central counterparty does not make counterparty risk disappear; rather, it centralizes and converts counterparty risk into different forms of financial risk

III. Unlike other financial institutions, the central counterparty cannot fail

- A. I only
- B. III only
- C. I and II
- D. I and III

58. Which of the following is the main source of liquidity risk in central counterparties?

- A. High operating costs
- B. Overinvestment in long-term assets
- C. Varying margin payments
- D. Overinvestment in derivatives assets

59. A foreign currency analyst based in Canada is forecasting the forward exchange rate CAD/USD. Currently, the spot exchange rate is CAD 1.21 per USD. Suppose that the interest rate in Canada is 1%, and the interest rate in the U.S. is 3%, which of the following is closest to the forward exchange rate of CAD per USD?

- A. CAD 1.1865 per USD
- B. CAD 1.2463 per USD
- C. CAD 1.2339 per USD
- D. CAD 1.2221 per USD

60. Consider the following information:

Currency	Libor (annualized)	Currency Combinations	Spot Rate
USD	0.30%	USD/EUR	1.6975
EUR	5.00%	JPY/EUR	0.0085
JPY	0.30%	JPY/USD	82.25

If the uncovered interest rate parity holds, what is the forward rate of JPY/EUR currency for one year?

- A. 0.0045
- B. 0.0089
- C. 0.0056
- D. 0.0065

61. Joachim Corporation issued four types of debt to raise the capital of USD 730 million. The debt includes USD 400 million raised through secured bonds, USD 110 million through subordinate bonds, USD 40 million through unsecured loans, and USD 180 million through debenture bonds. If the company defaults, the total value of the assets to be distributed would be USD 630 million.

The total claim that the subordinate bondholders would receive is closest to:

- A. USD 0
- B. USD 180 million
- C. USD 50 million
- D. USD 10 million

62. In a planned amortization class (PAC) collateralized mortgage obligation (CMO), when compared to the underlying mortgage-backed security, the planned amortization class (PAC) tranches have:

- A. Reduced contraction risk
- B. Reduced extension risk
- C. Both reduced contraction risk and reduced extension risk
- D. None of the above

63. Which of the following statements is incorrect regarding mortgage-back securities?

- A. Commercial mortgage-backed securities tend to be less volatile and complex compared to residential mortgage-backed securities
- B. The risk of prepayment typically occurs in declining rate environments
- C. When you invest in a mortgage-backed security, you are indirectly lending money to a home buyer or business
- D. None of the above

64. What is the most likely definition of a Special Purpose Vehicle (SPV)?

- A. A subsidiary company with an asset/liability structure and legal status that makes its obligations secure even if the parent company goes bankrupt
- B. A type of asset-backed security that is secured by a mortgage or collection of mortgages
- C. A structured financial product that pools together cash flow-generating assets and repackages this asset pool into discrete tranches that can be sold to investors
- D. A subsidiary that reduces the effects of prepayment risk – the risk that the principal amount will be paid ahead of schedule

65. Which of the following statements is correct regarding the value of a forward contract to a short party at expiration?

The value of the forward contract is:

- A. Equal to zero
- B. Equal to the value of the long party multiplied by -1
- C. Positive if the spot price of the underlying exceeds the forward price
- D. Either the full amount of the contract or zero

66. Jason Briggs purchased a 3-month call option by paying USD 0.08. The exercise price of the option is USD 1.32, while the underlying is priced at USD 1.35.

Is the option currently in-the-money, and at what price will break-even occur?

- A. In-the-money: No; Break-even Price: USD 1.27.
- B. In-the-money: Yes; Break-even Price: USD 1.40.
- C. In-the-money: Yes; Break-even Price: USD 1.35.
- D. In-the-money: No; Break-even Price: USD 1.35.

67. A European put option is selling for USD 4.00 with the underlying priced at USD 52. The exercise price is USD 50, and the underlying makes no cash payments during the life of the option. The risk-free rate is 6.0%, and the option expires in 120 days. A call with the same exercise price and expiry sells for USD 8.50.

This put option is most likely:

- A. Overvalued by USD 1.55
- B. Undervalued by USD 1.55
- C. Overvalued by USD 3.67
- D. Undervalued by USD 3.67

68. A trader applied a strategy where he purchased a European call option on the stock of KKL with a strike price of USD 29.50, and at the same time, sold a European call option on the same stock with a strike price of USD 34.89. Suppose that the final price of the stocks at expiration is USD 44, what are the name and the payoff of the strategy? Ignore the cost of the strategy.

- A. Name of the strategy: Bear call spread; Payoff: USD -9.11
- B. Name of the strategy: Bull call spread; Payoff: USD 9.11
- C. Name of the strategy: Bear call spread; Payoff: USD -5.39
- D. Name of the strategy: Bull call spread; Payoff: USD 5.39

69. Janet Marshall, a fixed investment manager at Premium S&B, has a short position in 10-year Treasury bond contracts with a USD 100,000 face value for each contract. The last quoted price of the contract is 98-30, while the accrued interest on the bond is USD 5.23 (for USD 100 face value). If the conversion factor for the deliverable bond under the contract is 1.375, then the cash price:

- A. Paid by the short party is USD 140.393
- B. Received by the short party is USD 140.393
- C. Paid by the short party is USD 141.269
- D. Received by the short party is USD 141.269

70. A deliverable bond has a maturity of 12 years and 7 months and pays a 6% coupon every six months. What is the conversion factor of the bond given that the discount rate is 7%, and the face value of the bond is USD 100 (maturity rounded to the nearest 3 months)?

- A. 0.9176
- B. 0.9172
- C. 0.9185
- D. None of the above

Valuation and Risk Models

71. Anna Frost is a junior options trader at an investment bank. She recently finalized a non-standardized options contract with a new client. The option had the following Greeks:

Delta	25
Gamma	100
Vega	100

Although Frost's supervisor does not mind bearing the risk of the changes in the price of the underlying, he is not comfortable with the Gamma and Vega risk exposures and asks Frost to hedge her position. At the moment, she has the following two options available on the market:

	Call Option 1	Call Option 2
Delta	0.55	0.85
Gamma	0.25	0.1
Vega	30	10

Which trades should Anna undertake to accomplish her supervisor's task?

- A. Buy 2,131 contracts of Call Option 1 and sell 5,639 contracts of Call Option 2
- B. Sell 1,980 contracts of Call Option 1 and buy 2,131 contracts of Call Option 2
- C. Buy 5,950 contracts of Call Option 1 and buy 1,980 contracts of Call Option 2
- D. Buy 1,980 contracts of Call Option 1 and sell 5,950 contracts of Call Option 2

72. An investment bank decided to open a position in the shares of ALNY's stock. The position will be opened by buying 5,000 out-of-the-money call options with a strike price of USD 350. At the moment, the stock is trading at USD 300. The annual volatility of the stock's return is 30%, and the delta of the option is 0.3. What is the daily VaR for this position at a 1% significance level, assuming 252 trading days in a year?

- A. USD 16,668
- B. USD 19,815
- C. USD 23,116
- D. USD 16,282

73. Ganz Hertz is the CEO of one of the biggest investment funds in Germany. Although the fund has always followed a balanced equity/fixed income allocation strategy, a recent rally in the real estate market prompted a reevaluation. Fritz Wolfgang, the chief investment officer, proposed to enlarge the fund's investment horizon and reallocate 10% of the fund's assets from bonds to real estate.

According to Wolfgang, the investment in a new asset class, which is not perfectly correlated with stocks and bonds, will not only help to increase returns but will also diversify risks and decrease the firm's VaR. To prove that Wolfgang is incorrect, Hertz quickly calculates VaR for the stock, bond, and real estate portfolios separately add them up (taking into account the proposed allocation proportions), and compares the final result with the VaR of the current portfolio. Wolfgang is disappointed with results because the VaR of the current portfolio turns out to be lower than the one calculated by Hertz.

Based on the information provided, which of the following statements is correct?

- A. The impact of the new asset class inclusion on portfolio VaR is correct, and the portfolio VaR calculation approach is also correct
- B. The impact of the new asset class inclusion on portfolio VaR is correct, while the portfolio VaR calculation approach is incorrect
- C. The impact of the new asset class inclusion on portfolio VaR is incorrect, while the portfolio VaR calculation approach is correct
- D. The impact of the new asset class inclusion on portfolio VaR is incorrect, and the portfolio VaR calculation approach is also incorrect

74. Which of the following statements is (are) accurate?

I. VaR is not a coherent risk measure while Expected Shortfall is

II. Although the calculation of Expected Shortfall is more difficult, it provides a narrower range of expected losses than VaR does under the same significance level

- A. Both I and II
- B. I only
- C. II only
- D. None of the above

75. Donat Bank from Slovenia recently started trading in FX options. Senior management wants to receive reports about risks arising from those trading activities and wants from its risk management unit explanations regarding the techniques used to measure risk. Which technique(s) is (are) most appropriate in this case?

- A. Delta normal
- B. Full revaluation
- C. Both A and B
- D. None of the above

76. Because of the recent volatility of interest rates, the CRO of a hedge fund proposes to completely eliminate the 5-year, 10-year, and 30-year exposure of the fixed income portfolio. The table below presents the key rate '01s of the current portfolio and corresponding hedging instruments:

Hedging Bonds	Key Rate '01 (per 100 face amount)		
	5-year	10-year	30-year
Bond 1	0.0175	-	-
Bond 2	0.0099	0.0250	-
Bond 3	0.0232	0.0314	0.0487
	Key Rate '01 (USD)		
	5-year	10-year	30-year
Fixed Income Portfolio	80.4895	193.4292	300.0000

What is the CRO's hedging transaction?

- A. Sell USD 357,000 of Bond 1; Sell USD 127,000 of Bond 2; Sell USD 616,000 of Bond 3
- B. Buy USD 270,000 of Bond 1; Sell USD 127,000 of Bond 2; Buy USD 616,000 of Bond 3
- C. Sell USD 270,000 of Bond 1; Buy USD 127,000 of Bond 2; Sell USD 616,000 of Bond 3
- D. Buy USD 357,000 of Bond 1; Sell USD 616,000 of Bond 3

77. The following table provides the initial price of a C-STRIP and its present value after application of a one basis point shift in four key rates.

	Value
Initial value	26.11485
2-year shift	26.11582
5-year shift	26.11778
10-year shift	26.13883
30-year shift	26.01772

The key rate duration with respect to the 30-year shift is closest to:

- A. 35.55
- B. 26.02
- C. 37.19
- D. 97.13

78. Banca Antiniana from Naples, Italy, needs to calculate the risk level of several bond portfolios. Particularly, the bank needs to measure and hedge the risk of portfolios containing both bonds and swaps. What are the instruments which will be used in the calculation?

- A. Partial '01s
- B. Forward-bucket '01s
- C. Both partial '01s and forward-bucket '01s
- D. None of the above

79. Joshua Manson, a bond trader at Capital Bank, has been asked to calculate the price of U.S. Treasury 4-5/8s of June 30, 2020. Joshua was provided with the following information:

Settlement Date	May 11, 2017
Coupon payment dates	June 30 and December 30
Yield to maturity	5.5%

Note that prices of Treasury bonds are calculated using the ACT/ACT day count convention.

What are the clean and dirty prices of the tranche?

- A. Clean price = 97.5035; Dirty price = 99.1808
- B. Clean price = 99.1808; Dirty price = 97.5035
- C. Clean price = 96.4752; Dirty price = 98.1524
- D. Clean price = 98.1524; Dirty price = 96.4752

80. Marmagne Bank from Bourges, France, needs to calculate the risk level of some bond portfolios. In this stage of the calculation, the Bank does not plan to hedge any of the portfolios but only to closely examine and understand the curve risks of the portfolios in question. What would be the instruments best used for the purpose of understanding the curve risks?

- A. Partial '01s
- B. Forward-bucket '01s
- C. Both partial '01s and forward-bucket '01s
- D. None of the above

81. A regional bank recently invested in a new bond-pricing software. To test it, the risk management team priced a bond at 3 different rates. The results are presented below:

Interest Rate	Price of bond
2.13%	USD 100.115
2.15%	USD 99.100
2.17%	USD 98.250

What is the estimate of the bond's convexity?

- A. 104.06
- B. 416.25
- C. 10,406.15
- D. 41,624.62

82. An employee at a large Wall Street bank is estimating the 5% VaR of a specific asset by using historical simulations to estimate the percentile of returns and using exponentially declining weights on past data. This methodology is most likely an example of:

- A. Parametric VaR estimating
- B. Nonparametric VaR estimating
- C. Implied Volatility VaR estimating
- D. Hybrid VaR estimating

83. Ben Thompson, FRM, is the senior risk manager at FTK bank. The bank has a significant proportion of its portfolio allocated to AAA Holdings. Thompson is concerned with the effect of a recent 5% drop in prices of AAA Holdings on the firm's portfolio VaR. To estimate the effect, he wants first to calculate the updated volatility of AAA Holdings using the GARCH(1,1) model. Thompson knows that the most recent and the long-term daily volatilities of holding are 2.7% and 2%, respectively. He also estimates the model's parameters as:

$$\alpha = 0.12; \beta = 0.8; \text{ and } \omega = 0.000032$$

What is the updated volatility estimate of AAA Holdings?

- A. 2.8%
- B. 2.9%
- C. 3.0%
- D. 4.0%

84. An analyst just finalized his presentation to the investment committee of an investment bank where he showed his calculation of the price of a 1-year European put option on stocks of one of the local companies using a binomial tree model. He used the following inputs:

Stock price	USD 100
Stock price annual volatility	30%
Option's strike price	USD 90
Risk-free rate	3%

A few hours later, he decided to revise the calculations and found a mistake in the calculation of the stock price annual volatility.

How will the price of the European put option change if the correct annual price volatility is 50% instead of 30%?

- A. +USD 8.80
- B. -USD 8.80
- C. +USD 5.25
- D. -USD 5.25

85. Futures on one of the Fortune 500 companies are correctly trading at USD 88 per contract. The volatility of these futures contracts is 21%. What is the price of a one-year European call option with a USD 85 strike price on this futures contract using the one-step binomial tree model if the risk-free rate is 3% per year?

- A. USD 11.43
- B. USD 11.86
- C. USD 12.21
- D. USD 23.56

86. A Treasury department analyst calculates that the open EUR currency position of Garden Bank is EUR 1,000,000. The Euro is currently worth USD 1.15, and the exchange rate has the annual volatility of 25%. The risk-free rates on USD and EUR are 2% and 0.5%, respectively. To hedge the position, the analyst suggests selling a 3-month call option with a strike price of 1.1.

What is the price of the European call option using a one-step binomial tree model?

- A. USD 0.0958
- B. USD 0.0978
- C. USD 0.0988
- D. USD 0.0998

87. Alina Mue is analyzing the performance of ABS stocks. Stocks are currently traded on the market at USD 120 and have an expected return of 25% per year, with the annual volatility of returns of 60%. Mue is requested to calculate a 95% confidence interval for the stock price in three months under the assumption of the lognormal distribution of returns.

Which of the following is closest to the confidence interval calculated by Mue?

- A. (USD 92.02; USD 150.61)
- B. (USD 67.83; USD 219.86)
- C. (USD 102.37; USD 145.68)
- D. (USD 60.70; USD 245.67)

88. Consider a European call option with the following parameters:

Strike price	USD 48
Expiration	6 months
Underlying's Price	USD 50
Annual volatility	25%

Assuming a risk-free annual rate of 8%, what is the probability that the option will be exercised in a risk-neutral world? (If required, use the table at the beginning of the document for statistical calculations.)

- A. 0.70
- B. 0.30
- C. 0.64
- D. 0.36

89. The risk manager of a large investment bank is reviewing the bank's investments in options contracts. He is particularly interested in call options contracts on shares of Hamilton Invest that the bank bought a few months ago. Hamilton Invest just unexpectedly announced that they would pay a USD 3 dividend per share in the sixth and twelfth months. The risk manager is concerned with the impact of dividends on the option's price. The risk-free rate is 5%, and the option has the following characteristics:

Strike price	USD 140
Expiration	13 months
Underlying's price	USD 151
Annual volatility	35%

By how much will the price of the option change after the announcement of the dividends? Assume that $N(d_1)$ before and after the announcement of the dividend is 0.7654 and $N(d_2)$ before and after the announcement of the dividend is 0.5489?

- A. The price of the option will increase by USD 3.32
- B. The price of the option will decrease by USD 3.32
- C. The price of the option will decrease by USD 3.82
- D. The price of the option will decrease by USD 4.42

90. A trader at an investment bank was requested to value non-standardized options on stocks of RNK Construction using the Black-Scholes-Merton model. Stocks are currently trading at USD 120 per share on the market, and the risk-free rate is 5%. The options contract has the following features:

Option type	European Put
Strike price	USD 100
Expiration	1 year

The trader calculates the following parameters for the model:

$N(d_1) = 0.7337$; and $N(d_2) = 0.6313$.

The price of the option is closest to:

- A. USD 2.88
- B. USD 3.01
- C. USD 3.12
- D. USD 3.47

91. Consider a call option on a non-dividend paying stock where the stock price is USD 79, the risk-free rate is 5%, the time to maturity is 1 month, and $N' = 0.3788$. By how much will a 1% increase in the volatility of the underlying change, the value of the call option?

- A. -USD 0.0249
- B. -0.0863
- C. +USD 0.0249
- D. +USD 0.0863

92. The Treasury Department of a regional bank recently got additional liquidity, amounting to USD 10,000,000. Gayane Gag, head of the treasury department, is interested in a T-Bond that is maturing at the end of September 2017. The tranche is paying a coupon rate of 2.5%. Although the tranche is not actively traded on the market, Gayane manages to collect the following information:

	Maturity	Coupon Rate	Price (per USD 100 face value)	Coupon frequency
Tranche 1	30/09/2018	1.5%	USD 97.54	Semiannual
Tranche 2	30/09/2018	2.5%	n/a	Semiannual
Tranche 3	30/09/2018	3%	USD 103.72	Semiannual

All three tranches have a semiannual frequency of coupon payments.

Based on the above information, what is the price of Tranche 2?

- A. USD 101.527
- B. USD 101.887
- C. USD 103.028
- D. USD 101.66

93. A risk manager prepares a presentation on the interest rate risk of the bank's bond portfolio. The table below shows the value of the portfolio in cases of key rate shifts and corresponding key rate '01s.

	Value	Key Rate '01
Initial Curve	900.525	
2-year Shift	898.950	1.5750
5-year Shift	900.302	0.2230
10-year Shift	889.112	11.4130
30-year Shift	?	-0.5980
Total		12.613

What are the portfolio's value and key rate duration for a 30-year shift?

- A. Portfolio's value = 901.123; Key rate duration = -6.641
- B. Portfolio's value = 901.123; Key rate duration = -6.641
- C. Portfolio's value = 899.927; Key rate duration = 135.090
- D. Portfolio's value = 899.927; Key rate duration = -9.119

94. A generalized autoregressive conditional-heteroskedastic (GARCH)(1,1) model has the following parameters:

$$\omega = 0.0005; \alpha = 0.01; \beta = 0.98$$

The implied long-run volatility level is closest to:

- A. 0.05
- B. 0.2236
- C. 0.0003
- D. 0.0159

95. A risk analyst uses the EWMA model with $\lambda = 0.85$ to carry out an update of correlation and covariance rates. On day $n - 1$, the observed percentage changes for variables X and Y are 5% and 3%, respectively. Historical data puts the correlation estimate between X and Y at 0.66 on day $n - 1$. If the estimated standard deviations on day $n - 1$ are 2.9% and 3.5% for X and Y , respectively, then which of the following is closest to the new estimate of the covariance between X and Y on day n ?

- A. 0.5612
- B. 0.0008
- C. 0.0011
- D. 0.0014

96. Almytra Bank approved a loan with a total commitment of USD 108 million. The bank disbursed an amount of USD 66 million, which is also the total outstanding loan amount. The credit risk department provided the following details with regard to the riskiness of the loan:

- The standard deviation of the probability of default and the loss rates are 3% and 11%, respectively
- The probability of default over the next year is 1.5%
- The loss rate if the customer defaults is 22 percent

What is the unexpected loss for the customer?

- A. USD 990,131.99
- B. USD 449,006.20
- C. USD 544,500
- D. USD 437,233.50

97. Austral-Asian Bank uses the advanced measurement approach to calculate the operational risk capital for regulatory purposes. Samuel Baker, a financial controller, working at the bank, analyzes the research reports on internal data and external data. He concludes the following from his analysis:

I. The loss frequency distribution must be specific to the bank and based on internal data and scenario analysis estimates

II. Historically, banks have done a much better job at documenting their operational losses than their credit risk losses

Which of these statements are incorrect?

- A. I only
- B. II only
- C. I and II
- D. None of the above

98. An investment bank has hired external consultants to review its stress testing procedures. After a month of thought investigations, the consultants came up with the following comments:

I. The bank has ensured an appropriate coverage in stress-testing activities, as the stress testing is applied to the business line, portfolio, and risk type levels, and not on an enterprise-wide basis.

II. Stress testing should be conducted over various relevant time horizons. The bank should concentrate not on conditions that may materialize in the near term but also on adverse situations that take longer to develop.

Which of the comments is (are) correct?

- A. Comment I only
- B. Comment II only
- C. Neither of the comments is correct
- D. Both comments are correct

99. A risk manager wants to stress a bank's USD 320 million real-estate loan portfolio. Although at the moment the average probability of default (PD) for the portfolio is pretty high (11.5%), the bank's management doesn't worry about the improvement of the credit quality of the portfolio as the collateral for the loans is a very expensive commercial property that always had high demand.

Based on the experience of 2008, the risk manager assumes a sequence of similar negative economic events that would not only increase the PD to 20% of the portfolio but also drastically decrease recovery rates on the loans to 40%. What is the real-estate portfolio's expected loss in the stressed scenarios?

- A. USD 10,800,000
- B. sUSD 14,720,000.
- C. USD 25,600,000.
- D. USD 38,400,000.

100. Which of the following statements regarding stress testing methodologies is incorrect?

- A. The impact of stress tests is evaluated against the economic profit or loss
- B. An appropriate stress testing framework should comprise a broad range of scenarios covering risks at different levels of granularity
- C. The stress testing program should explicitly cover complex and bespoke products such as securitized exposures
- D. A bank should enhance its stress testing methodologies to capture the effect of reputational risk